# UNRAMIFIED AT A POINT CHARACTERIZED BY COTANGENT SPACE, LOCAL EMBEDDING INTO AFFINE SPACE, AND LOCAL ISO OF THE DIAGONAL

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Source reference.* Original reference 02GF: sources/stacks-chow/tags.txt, source lines 3226–3226. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original definition 024N: sources/stacks-chow/tags.txt, source lines 2784–2784. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02G4: sources/stacks-chow/tags.txt, source lines 3215–3215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 039J: sources/stacks-chow/tags.txt, source lines 4274–4274. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 08S2: sources/stacks-chow/tags.txt, source lines 11115–11115. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01UZ: sources/stacks-chow/tags.txt, source lines 2428–2428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01UT: sources/stacks-chow/tags.txt, source lines 2422–2422. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01OW: sources/stacks-chow/tags.txt, source lines 2209–2209. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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*Date*: stacks-project:etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 00RT: sources/stacks-chow/tags.txt, source lines 1015–1015. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Theorem 0.1.** *Let $Y$ be a locally Noetherian scheme. Let $f : X \to Y$ be a morphism of schemes which is locally of finite type. Let $x$ be a point of $X$. The following are equivalent*

(1) *$f$ is unramified at $x$,*
(2) *the stalk $\Omega_{X/Y,x}$ of the module of relative differentials at $x$ is trivial,*
(3) *there exist open neighbourhoods $U$ of $x$ and $V$ of $f(x)$, and a commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow{\quad i \quad} & \mathbf{A}^n_V \\ & \searrow \quad \swarrow & \\ & V & \end{array}$$

*where $i$ is a closed immersion defined by a quasi-coherent sheaf of ideals $\mathcal{I}$ such that the differentials $dg$ for $g \in \mathcal{I}_{i(x)}$ generate $\Omega_{\mathbf{A}^n_V/V,i(x)}$, and*
(4) *the diagonal $\Delta_{X/Y} : X \to X \times_Y X$ is a local isomorphism at $x$.*

*Proof.* The equivalence of (1) and (2) is proved in Morphisms, Lemma 02GF.

If $f$ is unramified at $x$, then $f$ is unramified in an open neighbourhood of $x$; this does not follow immediately from Definition 024N of this chapter but it does follow from Morphisms, Definition 02G4 which we proved to be equivalent in Lemma 039J. Choose affine opens $V \subset Y$, $U \subset X$ with $f(U) \subset V$ and $x \in U$, such that $f$ is unramified on $U$, i.e., $f|_U : U \to V$ is unramified. By Morphisms, Lemma 02GE the morphism $U \to U \times_V U$ is an open immersion. This proves that (1) implies (4).

If $\Delta_{X/Y}$ is a local isomorphism at $x$, then $\Omega_{X/Y,x} = 0$ by Morphisms, Lemma 08S2. Hence we see that (4) implies (2). At this point we know that (1), (2) and (4) are all equivalent.

Assume (3). The assumption on the diagram combined with Morphisms, Lemma 01UZ show that $\Omega_{U/V,x} = 0$. Since $\Omega_{U/V,x} = \Omega_{X/Y,x}$ we conclude (2) holds.

Finally, assume that (2) holds. To prove (3) we may localize on $X$ and $Y$ and assume that $X$ and $Y$ are affine. Say $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$. The point $x \in X$ corresponds to a prime $\mathfrak{q} \subset B$. Our assumption is that $\Omega_{B/A,\mathfrak{q}} = 0$ (see Morphisms, Lemma 01UT for the relationship between differentials on schemes and modules of differentials in commutative algebra). Since $Y$ is locally Noetherian and $f$ locally of finite type we see that $A$ is Noetherian and $B \cong A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$, see Properties, Lemma 01OW and Morphisms, Lemma 01T2. In particular, $\Omega_{B/A}$ is a finite $B$-module. Hence we can find a single $g \in B$, $g \notin \mathfrak{q}$ such that the principal localization $(\Omega_{B/A})_g$ is zero. Hence after replacing $B$ by $B_g$ we see that $\Omega_{B/A} = 0$ (formation of modules of differentials commutes with localization, see Algebra, Lemma 00RT). This means that $\mathrm{d}(f_j)$ generate the kernel of the canonical map $\Omega_{A[x_1,\ldots,x_n]/A} \otimes_A B \to \Omega_{B/A}$. Thus the surjection $A[x_1, \ldots, x_n] \to B$ of $A$-algebras gives the commutative diagram of (3), and the theorem is proved. □